Competitive Analysis

Objective:

- To understand how other cloud/app services use & display consumption metrics

Project goal:

- To give customers a top level view to understand how much they are using and what they are being billed for before bill happens
- To give customers an opportunity to identify they are close to running up a bill
- Most important to include:
  - Breakdown of metrics (Data Transfer & Compute [GB/s and # of requests])
  - Actual number of usage that the user can see
  - Visual component
  - Links to documentation on billing & troubleshooting

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Executive Summary

Common Themes

1. Billing alerts & notifications sent to email or either directly on the dashboard
2. To show threshold distance, many companies use part of a whole
    a. Ex:
        i. Horizontal bar that represents 100 threshold, 30% is colored
        ii. Line at 100 with colored area underneath and bar graphs below

Takeaways

1. Include way user can troubleshoot - link to docs (?)
2. Billing alerts - way for user to set notifications at different levels within threshold
    a. Goal: To give customers an opportunity to identify they are close to running up a bill
3.

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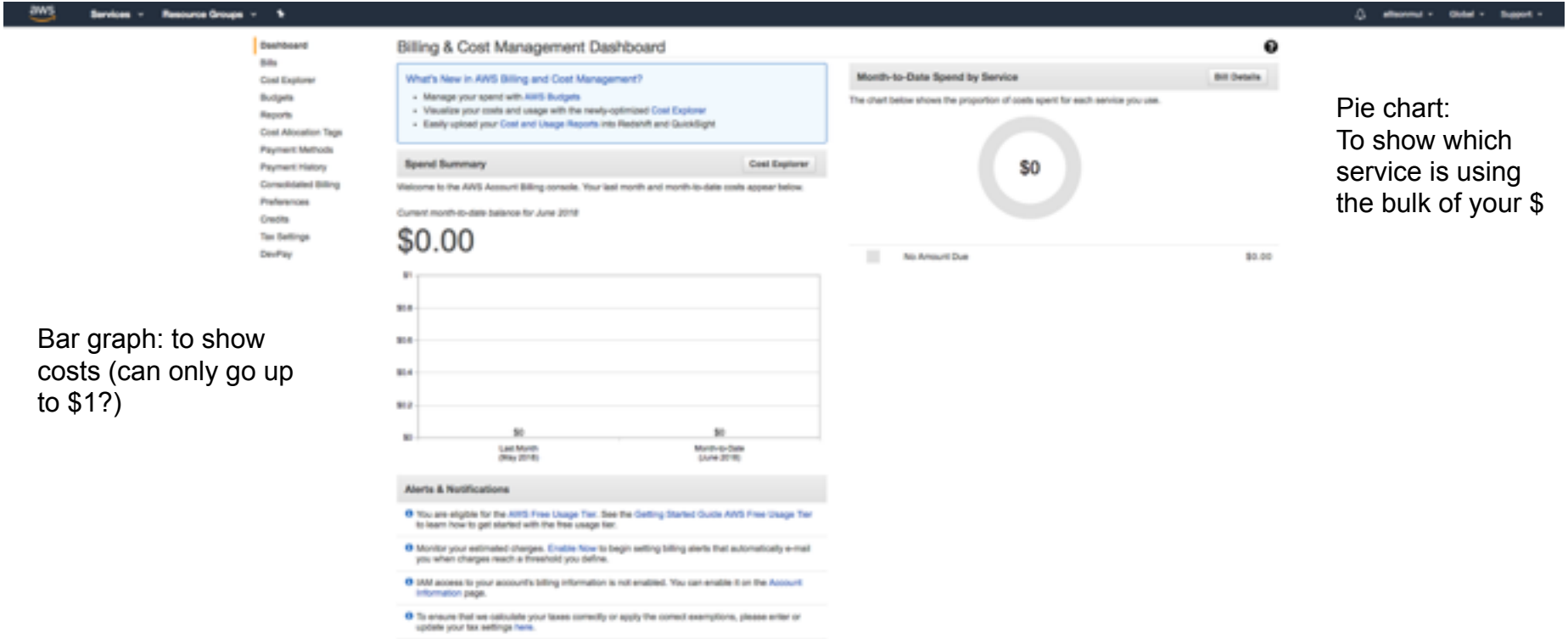
Executive Summary

Future Questions

1. How should forecasting look in the future?
2. Am I designing a section for a new page or section of ‘Getting Started’ page?

Amazon Web Services



Pie chart:
To show which service is using the bulk of your $

Bar graph: to show costs (can only go up to $1?)

Amazon Web Services

- https://console.aws.amazon.com/billing/home?#/
- Billing Alarms
  - Notifies user by email when you exceed the free tier limits or when forecasted to go over the free tier limits
- See usage with Top Free Tier Services by Usage Table - dashboard of Billing & Cost Management console
- AWS Cost and Usage Reports
  - AWS Cost and Usage reports provide access to detailed data, enabling you to better analyze and understand your AWS costs as well as the specific product offerings and usage amounts underlying those costs. You can customize the content and delivery of your reports and manage them from the reports dashboard.
- Cost Explorer - reporting, analytics, and visualization capabilities to track and manage costs

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Amazon Web Services: Takeaways

- Billing alarms could be something useful
  - *Thought: This is how user sees if usage will exceed cap*
    - *How to troubleshoot?*
- AWS takes billing as a whole & uses it as a whole different section
- Charts
  - Bar chart to show consumption
  - Pie chart to show service cost allocation
    - *Possible con: pie charts have been known to be ambiguous in some ways b/c user may need reference points or hard to distinguish b/w 4% and 6%- would need extra labels to show the exact %*
      - *Is % helpful?*
      - *TBH I don't really like pie charts*
    - *Thought: how else would you show a "part of a whole"?*

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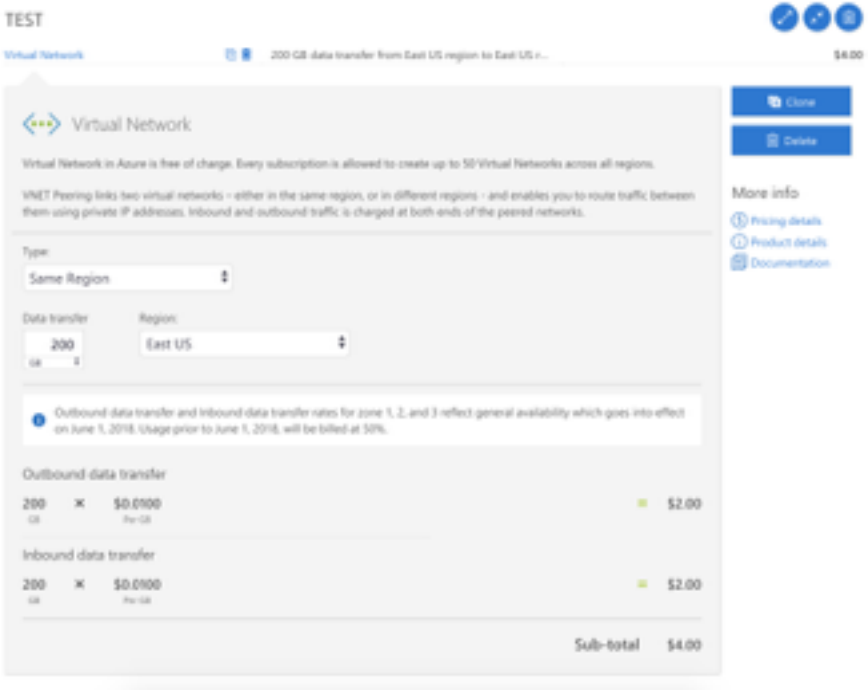
Microsoft Azure

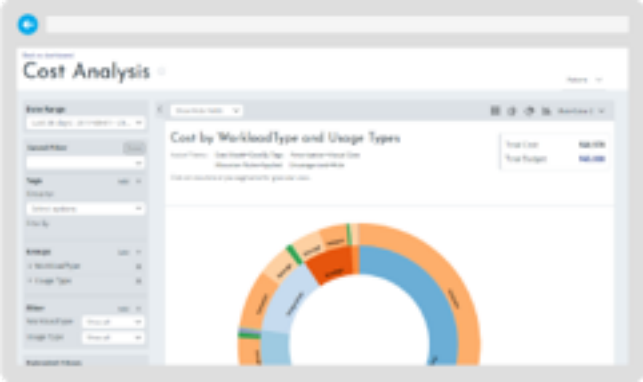
Pricing Calculator



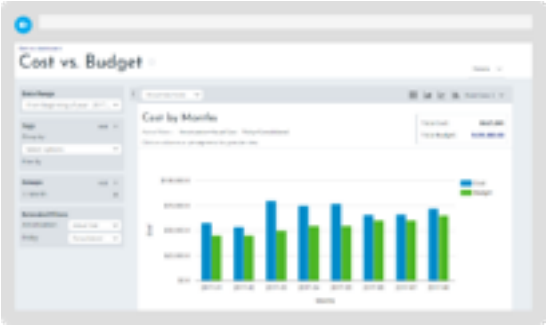
Microsoft Azure

Cost Management System
https://azure.microsoft.com/en-us/services/cost-management/

- Historical data for forecasting cloud usage & expenditure
- Continuously monitors consumption & cost trends
- Improves organizational accountability by allocating cloud costs to business units & projects
    - How users tell what service is using how much $$



Pie chart: visualize cloud usage and costs - services & $



Bar chart: actual usage vs budget & forecasts future usage

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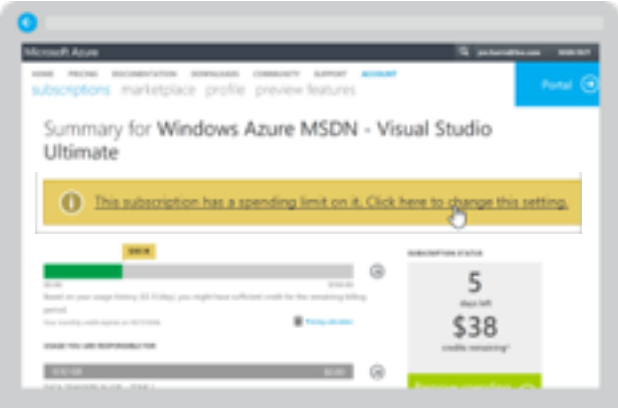
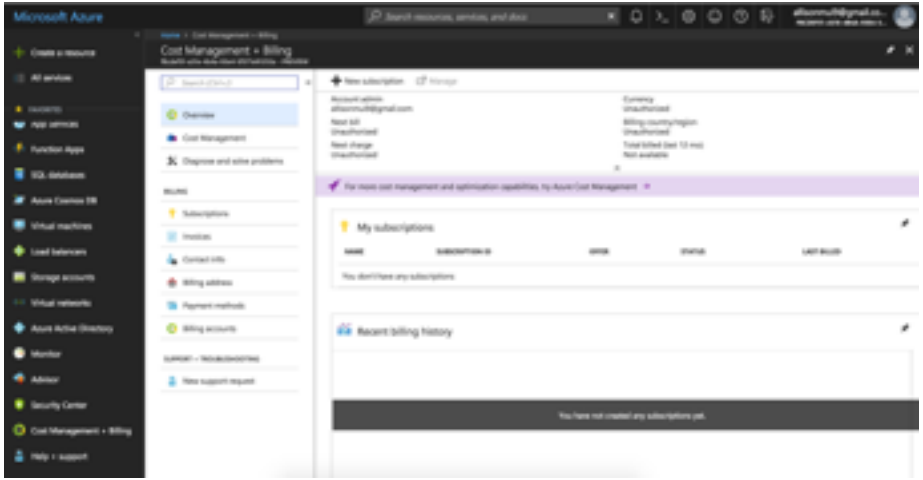
Billing Portal
https://docs.microsoft.com/en-us/azure/billing/billing-getting-started



Sorry so blurry ~
Troubleshoot action button
Horizontal bar graph - usage out of until free tier is up



How dashboard looks with no info

Included to see layout of dashboard

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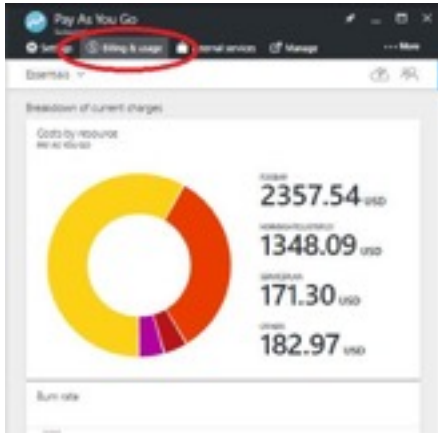
Billing Portal
https://docs.microsoft.com/en-us/azure/billing/billing-getting-started

- *Also offers Cost analysis for billing history*
  - Cost invoice which shows breakdown of total cost (from all months) by invoice
  - Not pictured but just looks like a regular table
- Dashboard is vertically one after the other or horizontally next to each other based on window size

Pie chart - breakdown of services
Line graph (with area underneath) - consumption and projected consumption



Microsoft Azure

- Pricing calculator
  - Use this before you pick a service to run on Azure - even before signing up for an account
  - For customers to see if they like the price before committing to it
- Billing portal
  - Displays: cost by resources, burn rate, cost projection, billing history, cost management system
- Automatic email billing alerts
  - Similar to what AWS does

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Microsoft Azure: Takeaways

- Pricing calculator
  - Use this before you pick a service to run on Azure - even before signing up for an account
  - For customers to see if they like the price before committing to it
- Billing portal
  - Displays: cost by resources, burn rate, cost projection, billing history, cost management system
    - *Thoughts:*
      - *Pie charts again - is there a better way to represent a resources breakdown?*
      - *Line Graph with Area below for projections could be interesting.*
        - *In our case: how would it show how much the user has before reaching the end of the free tier?*
        - *Maybe not the graph for us*
      - *Has action to troubleshoot cost - for us: simple link to Docs?*
- Automatic email billing alerts
  - Similar to what AWS does

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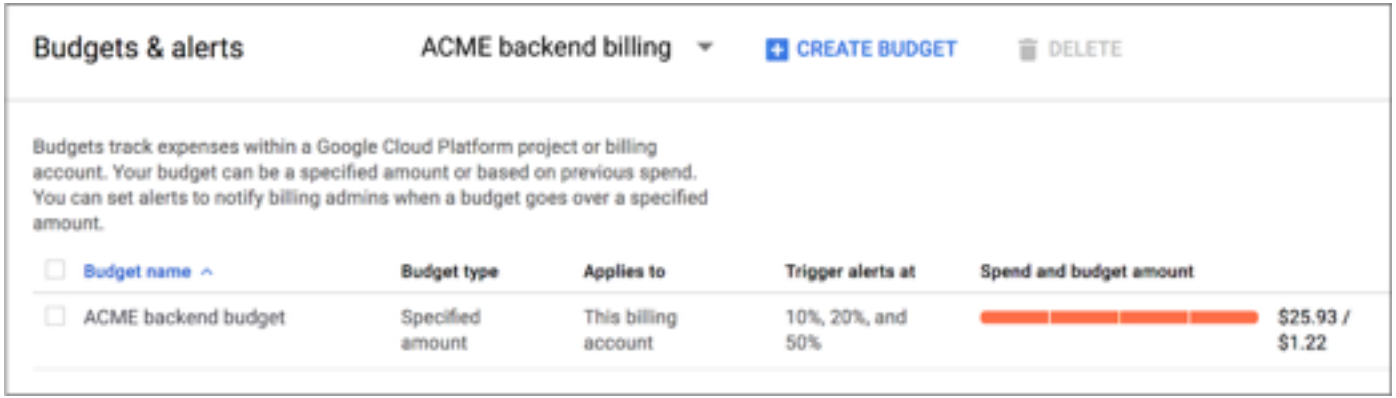
Google Cloud Platform

- https://cloud.google.com/billing/docs/how-to/reports
- Billing alerts and notifications setup
- Line graph to show cost and payment history
- Picture below shows budget - part by part horizontal bar graph



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Google Cloud Platform: Takeaways

- *Thought: Trigger alerts are helpful. So we also give users notice, not only, when they are on the brink of going over free tier, but also in incremental steps.*
  - *More time to troubleshoot*
  - *GCP uses 10, 20, 50%*

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Heroku & Datadog

Heroku

- https://devcenter.heroku.com/articles/language-runtime-metrics#general-information
- Pictured Heroku's sample Language Runtime Metrics
- Written -
  - How much memory used last time, max memory, and average
- Chart -
  - Shows usage with area below

Datadog

- Simple bar graph to show total # of requests

Heroku Language Runtime Metrics

Datadog total # of requests

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Heroku & Datadog: Takeaways

- *Thoughts for Heroku:*
  - *I like the quick stats on what the usage was latest and max*
    - *Could be helpful to also include how much user used in total, not just the latest usage*
  - *Chart shows usage for 24 hour period*
    - *I **think** we would want to include a month period b/c*
      - *To represent the billing cycle*
      - *Free tier resets every month*

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